

How to train fascia in soccer

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Positive and negative influences on the myofascial system in soccer

Excellence in any sport demands a sport-specific level of physical conditioning, combined with corresponding technical skill, to ensure mastery of sport-specific patterns of movement, as well as the tactical insights needed to engage in competitive play.

Soccer, in particular, is characterized by a wide range of stereotypical patterns of movement that are sport- or discipline- specific. Related sports like American Football or Australian Rules Football show comparable stereotypical kicking movements. If these patterns of movement are performed cumulatively in sufficient number, over a prolonged period of time, it is reasonable to expect that these football-specific movement stimuli will provoke reactions that manifest themselves as adaptations of the particular biological structures involved (joints, ligaments, neuromeningeal and myofascial structures) to permit adequate “processing” of the incident stresses and loads (Chapters 4 and 5). Over the past 20 years, in the course of delivering medical care to soccer players at virtually every performance level from amateurs to professionals who represent their countries internationally, we have empirically identified a wide diversity of changes to which soccer players are prone. In this context, because it involves the side-specific dominance of a kicking leg and a support leg,

soccer is also characterized by corresponding asymmetries in adaptations and changes, particularly with regard to the myofascial system.

These adaptations maximize the quality of the sport-specific patterns of movement and thus serve to enhance the individual's performance in their particular sport. However, they are also often the cause of changes in muscular stress patterns within the sport and may, in certain circumstances, lead to abnormal loading or overloading of the body structures involved. Knowledge of the soccer-specific changes affecting the musculoskeletal, neuromuscular and myofascial systems enables coaches and the medical team to assess any structural and functional implications more easily, and lays the groundwork so that athletes can engage in structure-specific preparation and recovery routines. The information below is intended to raise awareness among coaches and the medical team concerning the existence of soccer-specific adaptations and to ensure these phenomena receive proper attention.

Soccer-specific changes and adaptations involving the musculoskeletal system

The following sections outline typical soccer-specific adaptations, with a special focus on myofascial changes which are encountered repeatedly in practice, even in the absence of

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injuries. A correspondingly high probable incidence of such changes must be anticipated in the active soccer player (that may also persist for years afterwards), and it is important to bear this in mind when preparing for both training and competitive matches. The following categories of change figure prominently in soccer and deserve attention.

Changes in the kicking leg due to ball contact

By definition, soccer entails a varying number of contacts with the ball.

In this process, the mechanical stresses associated with ball contact, if generated in sufficient number and magnitude, provoke changes in the biological structures. During the course of evolution, nature has developed our musculoskeletal system, and especially our lower limbs as components of the pelvic-leg axis, specifically for locomotion, walking and running (Chapters 29 and 30). Our feet, with their longitudinal and transverse arches, are ingeniously designed to cushion the impact of our body mass with each step, and to propel our bodies forward in the terminal phase of the gait cycle. Upon contact with the ball, a force is generated over a short period of time that is precisely opposed to the arched construction of the foot, giving rise to corresponding mechanical forces and stresses. With each individual kicking action, the mechanical reaction forces released by the ball mass are of a magnitude that remains well within the physiological range and generally does not exceed the stress tolerance of the structures involved. However, if this action is repeated a sufficient number of times over a prolonged period, possibly many years, the resultant stimuli then act as a form of micro-trauma, eventually leading to changes in the musculoskeletal system. Such changes have

been postulated and discussed since the mid-1980s, for example, by Hess (1985), and by Lees and Nolan (1998).

To ensure appropriate preparation for the sudden and brief tensile stresses generated by ball contact, strengthening occurs at the insertion sites of the talonavicular ligament, which is placed under tensile stress by the action of kicking. The increased numbers of more strongly developed Sharpey's fibers take up an increased amount of space and may then frequently be visible radiologically as a talar beak and/or tibial peak, a phenomenon that results in reduced dorsiflexion mobility at the ankle joint. Furthermore, the majority of soccer players have a flatter medial longitudinal arch (1–3 mm) in the kicking leg compared with the support leg, which consequently provokes an ascending chain of causes and effects with a subtly enhanced tibial internal rotation onto the navicular bone, with corresponding effects on the knee joint in the form of an increased “dynamic genu valgum” during the standing phase.

Alongside this and as a result of these direct changes to the ankle joint in response to kicking movements, soccer players are also likely to develop muscle changes that are characterized by varying degrees of right-left asymmetry (support leg vs. kicking leg). The kicking movement of the leg axis in question represents an “open kinetic chain” type of load in which the foot is moved with maximum forward velocity (moving point) and the hip remains relatively stationary (fixed point). At the same time, every kicking movement necessarily imposes a “closed kinetic chain” type of load on the non-kicking side. In this case, the non-kicking foot is planted on the ground (fixed point), while the structures above it throughout the pelvic-leg axis and torso are in motion (moving point) and

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need to be adequately stabilized against gravity by a complex set of coordination mechanisms. Similarly varied neuromuscular control actions then create the basis for long-term muscular adaptations to these soccer-specific movement patterns. In the long term, the active musculoskeletal system, as well as the myofascial system, may be assumed to adapt progressively to the characteristic movements that they are called upon to perform, and to the loads associated with those movements as they develop an optimized muscular response.

Reports describe muscle differences between the support leg in relation to the kicking leg (Ekstrand et al., 1983; Knebel et al., 1988). Kicking the ball is a multiple-joint movement, in which an obviously explosive extension movement at the knee is combined with active flexion of the hip and extension (plantar flexion) of the foot at the ankle joint. In general, these authors describe both an increased maximum strength capacity and an increased striking force during quadriceps extension on the kicking-leg side, accompanied by increased maximum strength and striking force of the knee flexors on the support-leg side (Figure 45.1).

Our own empirical observations on the degree of quadriceps muscle development in soccer players suggest additional neurophysiological aspects and considerations relating to long-term functional adaptations. Although soccer players have greater quadriceps strength on the kicking-leg side than on the support-leg side, examination of the thighs in most players shows that the thigh circumference tends to be slightly reduced in the area where the vastus medialis muscle is most fully developed. Evidently the variable “muscular configuration” of the quadriceps is an adaptive response of the musculoskeletal system to years of locally varying, stereotypical functional demands (the kicking leg with its

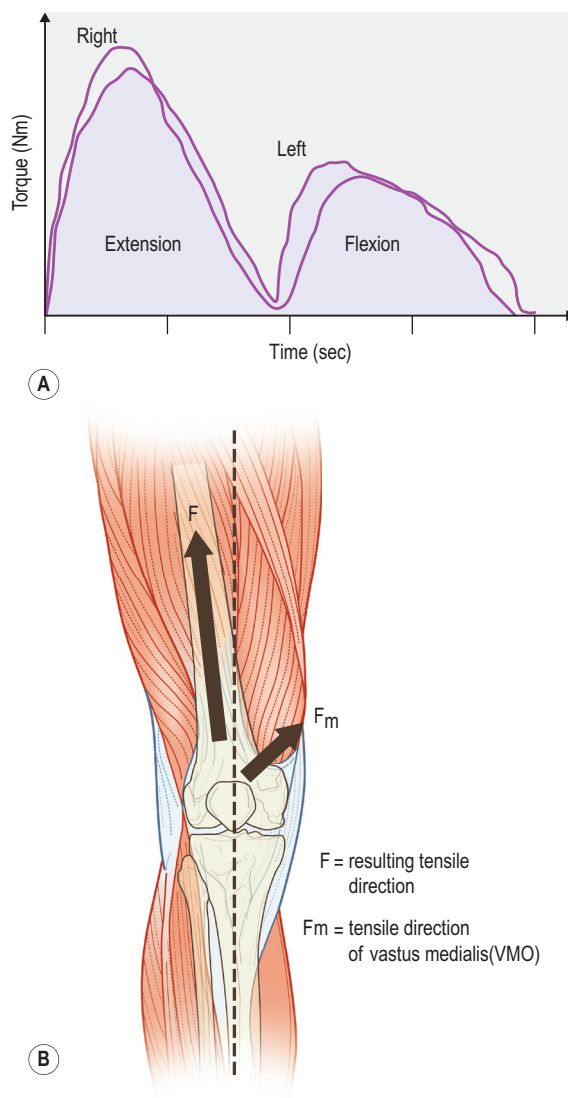


Figure 45.1 Agonist-antagonist relationship in the muscles of the knee joint

numerous open kinetic-chain loads, in contrast to the support leg with its closed kinetic-chain loads). It is not uncommon for there to be clearly identifiable volume deficiencies in the territory of the vastus medialis muscle of the kicking leg in some players.

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From a neurophysiological standpoint, a “chronic” vastus medialis deficiency in the kicking leg can be explained by the absence of any need during open kinetic-chain activity to stabilize the knee joint in terms of tibial rotation relative to the femur in response to gravitational effects. The long-term change in the innervation pattern gradually results in adaptation that is optimized for kicking the ball. The other side of the coin is that this process alters the relative contributions of the individual quadriceps muscles to the resultant quadriceps force. Deficiency of the vastus medialis tends to lateralize the quadriceps pull on the patella, thereby altering the kinematics of the femoropatellar joint. By contrast, the loads on the support leg during running and sprinting tend not to alter the physiological pattern of intra-articular kinematics. Consequently, degenerative changes in the femoropatellar joint are detected far more commonly on the kicking-leg side than on the support-leg side.

Support leg changes due to kicking technique

The changes in the kicking leg, described above, suggest that the contralateral support leg is obviously exposed to different loads during the kicking of a soccer ball. It is interesting that all soccer players, virtually irrespective of their performance level, tend to place their support leg in a very precise position when kicking the ball, with the instep or with the inside/outside of the foot. This results in a highly consistent pattern of stereotypical mechanical loads acting on the musculoskeletal structures. To ensure successful ball acceleration by following through with the kicking leg to transfer momentum effectively to the ball, the support leg must be planted correctly next to the ball on the ground. The following

initial observations are important in this regard (Hoffmann, 1984):

- As far as possible, soccer players plant their support leg next to the ball with remarkable consistency and precision. Tests have shown that intra-individual differences from one ball contact to the next are less than 1 cm (Hoffmann, 1984).
- Soccer players plant their support leg level with the ball (relative to the frontal plane).
- As the foot is planted on the ground, the body’s center of gravity shifts outward towards the support leg, usually moving beyond the left knee or even further laterally.
- The lateral distance of the support leg from the ball can vary markedly from one player to the next. Despite these pronounced differences, however, the individual solutions and movement patterns are performed with great precision (intra-individual consistency). However, the further the support leg is placed away from the ball, the greater the lateral shift in the body’s center of gravity. The joints along the pelvic-leg axis of the support leg have to stabilize and compensate for this position and this will then culminate in corresponding adaptive changes over time. These side-specific changes are most clearly identified in the ankle joint. The greater the lateralization of the pelvic-leg axis, the greater the lateral and shear forces acting on the joints of the foot. Even in the absence of injuries and/or trauma, these forces have the potential to induce long-term adaptations.

Adaptations of the lumbopelvic-hip region

The muscular adaptations, described above, which occur on the support-leg and kicking-leg

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sides in response to the demands of soccer also bring in their wake long-term changes in the entire lumbopelvic-hip complex. Our own gait analyses (kinematic, dynamic and palpatory findings) together with manual therapy assessments have highlighted the following aspects: the dominance of the powerful quadriceps and hip flexors (especially the iliopsoas muscle) on the kicking-leg side causes the pelvis to tilt posteriorly on that side (posterior pelvic tilt with inflare component). This, in turn, may cause an anterior pelvic tilt (relatively upright ilium with outflare component) to develop on the contralateral support-leg side in an attempt to stabilize the body's center of gravity in the long term. Additionally, these changes are often accompanied by a decreased range of motion in the sacroiliac joint on the kicking-leg side. This asymmetrical range of motion, combined with torsion of the hips, seems to produce apparent lengthening of the support-leg axis and leads to functional pelvic

obliquity. Furthermore, the new stress patterns are transmitted to the structures of the lumbar spine. As a result of the posterior pelvic tilt on the kicking-leg side, physical examination of soccer players often reveals that the lumbar spine is rotated posteriorly to the right due to increased tension on the iliolumbar ligaments.

Implications for the myofascial system

To summarize, soccer players can be expected to exhibit changes and side-specific adaptations affecting all the joints and ligaments of the lower limb, and hence the associated musculoskeletal structures together with the entire functional unit of the pelvic-leg axis with its associated myofascial chains. Varying in degree between different players, and depending on individual predisposition and the duration (years of training) and quantity (scope of training) of the loads and stereotypical movement patterns, these changes may become a source of problems.

Table 45.1 Muscles of the myofascial inflare and outflare chains

Adapted from Meert, 2003

Myofascial inflare chain Torsion + flexion (e.g. left-sided chain)	Myofascial outflare chain (Torsion + extension (e.g. left-sided chain))
Rectus capitis lateralis muscle, left	Obliquus capitis muscles, left longissimus capitis muscle, left trapezius muscle (descending part), left
Scalene muscles, left sternocleidomastoid muscle, left	Levator scapulae muscle, left
Major and minor pectoralis muscles, left	External rotators, left shoulder superior posterior serratus muscle, left
Intercostal muscles, left	Inferior posterior serratus muscle, left
External oblique abdominal muscle, left internal oblique abdominal muscle, right	Diaphragm (transverse separating structure) thoracolumbar fascia, quadratus lumborum muscle, left
Diaphragm (transverse separating structure)	Psoas muscle, right
Iliopsoas muscle, right	Gluteal musculature, right; tensor fasciae latae muscle, right
Obturator muscles, right adductor muscles, right vastus medialis muscle, right	Vastus lateralis muscle, right
Peroneal musculature, right	Sartorius muscle, right mediocrural musculature, right
Plantar fascia	Plantar fascia

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In consequence, some rethinking of priorities will be necessary, both when preparing for training/competitive matches and during the recovery (cool down) phase to optimize regeneration processes. Furthermore, alongside the classic and now conventional strategies (with their focus on joints, ligaments and the musculoskeletal system), consideration should also be given to the myofascia and to relevant exercises that will enable such considerations to be translated into practice. In this context, side-specific connective-tissue properties may also manifest themselves as a result of adaptation to the stereotypical stresses of training/competitive matches. Both connective tissue properties are often found to be reduced on the kicking-leg side. Both the long anterior myofascial chain (components still missing here) and the diagonal anterior myofascial chain (like *m. adductor longus* and *m. rectus abdominis*) are found to be less taut and, because of this reduced firmness, paradoxically

less elastic. The three-dimensional architecture of collagen “network structures” could therefore be altered in terms of firmness and elasticity (Chapter 8).

The lateral fascia (fascia lata and tractus iliotibialis) of the thigh is also more clearly pronounced with increased firmness. A dominant straight posterior myofascial chain and crossed posterior myofascial chain are found on the support-leg side (Figure 45.2).

Building up optimized fascial structures/networks in the medium and long term

On the basis of the anatomical, biomechanical and physiological principles previously outlined, we can formulate the following general pragmatic “fascial training principles” for the medium and long-term:

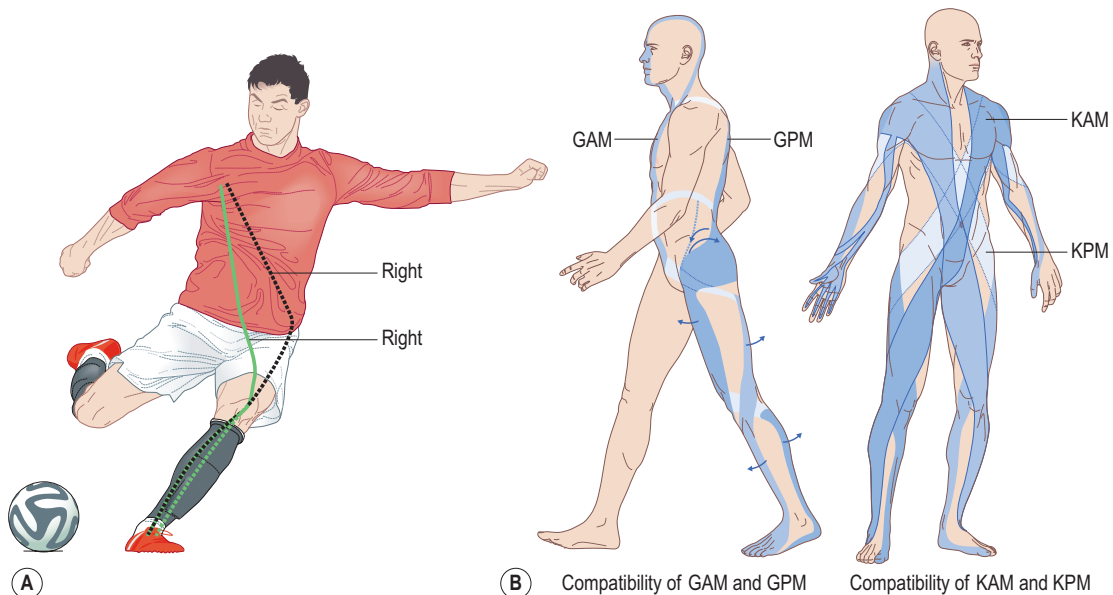


Figure 45.2 Soccer-specific stereotypical movements with associated myofascial chains

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Table 45.2 Fascial training principles

	Fascial tissue stretch	Fascial tissue elasticity	Fascial tissue tension
Training objective	- Increased synthetic activities of fibroblasts - Alignment of collagen fiber structure - Optimized nourishment of collagen fibers	- Optimization of strength development specifically related to the type of sport - Development of jumping power specific to the type of sport - Economization of energy metabolism	- Improvement of 3D-musculoskeletal stability work in the tensegral system - Formation of myofibroblasts in musculoskeletal transition
Training method	- Multidirectional stretches of myofascial chains across several joints - Self-myofascial release/foam rolling - Plyometric exercises (focus tendon energy storage via stretch-shortening cycles)	- Plyometric exercises (focus muscle-tendon connection via muscular force training effects) - Exercise variants with activation of the stretch-shortening cycle	- Whole-body exercises with high tension of the musculoskeletal system - Exercises with >65% of current maximum strength rating

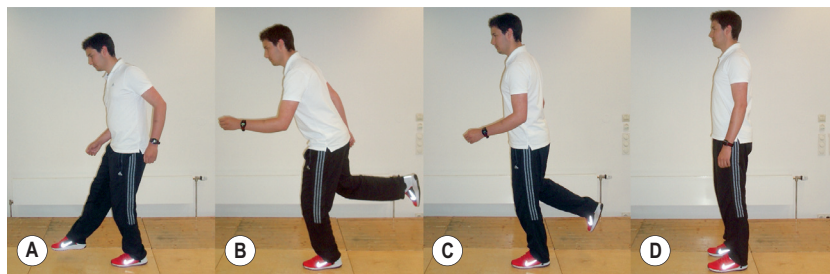


Figure 45.3 Soccer-specific complex stabilization exercise with contralateral trunk rotation and reversal of support/kicking leg function

Exercises and forms of movement, which by and large are focused on the turning points of movement that are specific to the type of sport (such as the kicking movements in soccer), where the agonists operating in synergy and the stretched antagonists meet, demand a virtually isometric tension of the contractile elements. This enables further small movements with directional change via the fascial (elastic) elements, as well as a developmental and directional stimulus. Preparatory counter-movements of this type, close to the turnaround points of movements, may be combined with a variety of rotational components (3D arrangement of the fascial network) (Chapter 40).

Taking account of the myofascial system when preparing for training and competition

Physiotherapeutic modalities

The soccer-specific profile of requirements and adaptations of the locomotor system should be taken into consideration in all preparation for training/competition. Although no reliable scientific evidence exists for the protective effectiveness of most of the following specific training recommendations, these recommendations reflect the current thinking and observations of our group and associated colleagues based on

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Figure 45.4 Mobilization of the iliopsoas muscle



Figure 45.5 Mobilization of the hamstrings

our extensive work with professional soccer players. The following two mobilizations may be performed for the prevention of dysfunctions and/or dysbalances.

- in the case of certain restrictions, irritate the structures of the affected pelvis-leg axis to a major extent and may thereby increase the risk of injury.

The iliopsoas muscle

- is connected to the respiratory diaphragm (Prometheus, 2005) and therefore, its proper function is important for breathing
- is “connected” with all of the discs in the lumbar spine
- surrounds the lumbar plexus of the motor and, among others, sensory of the quadriceps muscle and the knee joint
- contributes to the determination of stride length.

The hamstrings

- connect the pelvis to the lower extremity
- “interact” with the menisci (Barral and Croibier, 2013)
- “surround” the ischiatic nerve and its end-branches, the tibial and peroneal nerves

Preparing for training and competition

Passive modalities: Elastic taping and kinesiotaping

In preparing for training/competitive matches, elastic tapes and kinesiotapes may be used to stabilize joints. Functional tapes replace and supplement ligamentous and osseous joint-stabilizing factors and produce tensile and relaxing effects on the skin and subcutaneous receptors, thus selectively facilitating or inhibiting muscle tone/activity (neurophysiological and myotensive/myofascial improvement of joint stability via muscular joint stabilization factors). See Eder and Mommsen (2007), and Mommsen et al. (2007), for further information on these techniques.

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Active modalities

The purpose of all preparatory modalities/activities in this context can be subsumed under the following definition: optimal psychological and physical preparation for subsequent soccer-specific psychological and physical demands to achieve the current best possible level of performance.

In line with ideas originally outlined by Schlumberger (Eder and Hoffmann, 2006), these active modalities comprise, in general, the following key areas:

- Boosting physiological metabolic activities using moderate running with varying intensity to raise core body temperature, increase general metabolic activities, and optimize neuromuscular control processes.
- Structure-specific preparation of the musculoskeletal system for soccer-specific

movement sequences utilizing dynamic stretching and taking account of myofascial structures (Chapter 40).

- Preparation for soccer-specific stereotypical movement patterns with progressively increasing complexity and intensity. In particular, soccer-specific cutting movements with directional changes of varying intensity should be integrated in this process, as well as phases of acceleration and deceleration, initially without the ball, and then as preparation advances, receiving the ball and running with it, as well as situation-specific movement sequences (corners, free kick variations, etc.).

In terms of structure-specific preparation of the musculoskeletal system, in particular, the widespread practice of static stretching immediately prior to the stresses of training/competitive matches should be discarded

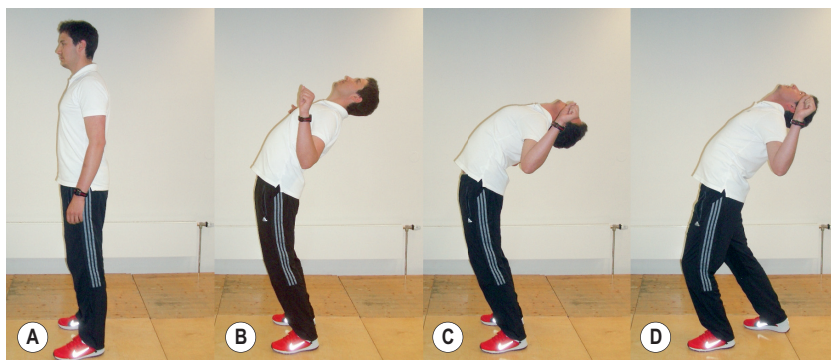


Figure 45.6 Stretching the straight anterior myofascial chain to prevent entrapment

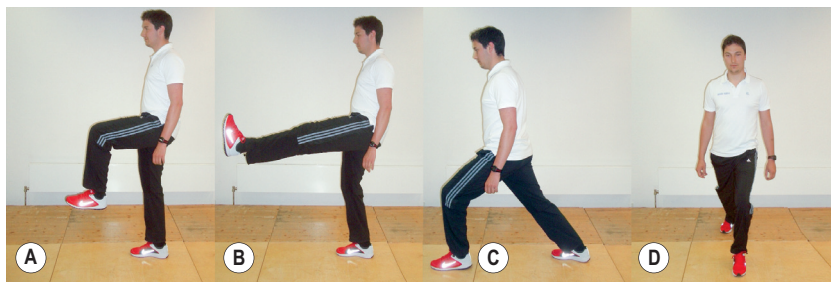


Figure 45.7 Example to illustrate dynamic stretching: hurdle walk

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(Chapter 10). Similarly, we propose that the exercise forms involving preparatory counter-movement and rapid bouncing as a type of “dynamic stretching” to shape and develop the fascial network are less suitable (if not actually unsuitable) as immediate preparation for the stresses of training/competitive matches. Instead, the suitable approach involves “dynamic stretching” that, as well as catering for the myotensive (muscular contractile) and neuromuscular (proprioceptive) movement prerequisites, also prepares the myofascial structures for the stresses and loads that are to follow.

In our experience, plyometric exercises (Chapter 25), initially with a unidirectional character and then building up with a multidirectional three-dimensional character (linear and lateral as well as rotatory), have proved themselves useful for activating joint-specific capsular receptors to round off structure-specific preparation, and can then transition into soccer-specific stereotypical movement patterns with and without the ball (Chapter 24).

Taking account of the myofascial system during recovery after the stresses of training/competitive matches

Generally recognized measures, after the stresses of training and competitive matches, for the timely restoration and recovery of physical performance, have the following objectives:

- Replenishment of exercise-induced fluid and energy deficits (for example, muscle glycogen stores).
- Optimization of physiological metabolic status (supply and removal in the context of biochemical regeneration processes) by promoting the circulation using moderate, low-intensity aerobic (alactic) movement

types incorporating exercises that are not soccer-specific, such as cycling or aqua-jogging (characterized by reduced intra-articular load stresses on the joints).

- Possibly, physiotherapy to monitor the status of the joints and ligaments of the lower limb and spinal column.

These activities should be supplemented with special reference to the fascial network. To release fascial adhesions and resolve swellings that have developed following the stresses of training/competitive matches, self-myofascial techniques can be integrated into the regeneration strategy, thereby promoting dynamic hydration of the collagenous ground substance (although these effects have not been confirmed by research so far) (Chapter 27). Notwithstanding, foam rollers may be suitable for this purpose. The locally limited contact surface of these rollers permits a slow gradual rolling movement via the gravitational application of the athlete’s bodyweight and so encourages a continuous further rolling movement along the course of the affected fascia. Here, too, in the specific context of soccer, it is important to direct such efforts preferentially to the most commonly affected fascia. In our experience these are the lateral structures of the kicking leg due to soccer-specific adaptations.

Conclusion

More recent scientific insights have brought about a paradigm shift in our understanding and appreciation of the function and importance of the fascial network. Alongside the classic targets of training, such as musculoskeletal function and performance, cardiopulmonary function and performance, and neuromuscular coordination and control, this radical change of perspective highlights the imperative to revise and re-orientate training objectives, and their

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Figure 45.8 Myofascial auto-release techniques for the dorsolateral fascia of the kicking-leg thigh

implementation and achievement, by adopting appropriate training modalities. In terms of fascial training, this requires measures and methods to develop and optimize a fascial network, as well as strategies to integrate fascial aspects

into direct preparation for and recovery from the stresses of training/competitive matches, thus ensuring that the fruits of modern fascia research are incorporated into the soccer-specific training process.

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